Independent source review: two finds and distinct-root union

The reviewed word composition follows the retained native reduced-label, attachment-orientation and anchor-update formulas under its stated premises. All 120 entries of its archived manifest matched current bytes. The three adapted modules differ from their originals only by the documented decoder/import renames. The retained readback contains exactly the expected seventeen names, no duplicates and no sorryAx. These are archival consistency observations; no fresh Lean process ran.

Scope and independence

Reviewer renew_materializer did not author the reviewed union proof. Producer summaries and the inherited coordinator history were already visible, so this is non-author source review, not blind derivation. The prior and attacks were recorded before reading detailed proof bodies. The review-evidence formal workflow was consulted. Its fresh-kernel-replay requirement was not performed because this assignment explicitly prohibits compilation and requests preservation of existing receipts. Machine status for this review is not_verified; archival producer status remains passed. Source fidelity is faithful_scoped for the enumerated conditional composition. Overall formal review outcome remains inconclusive pending independently admitted replay. No official hypothesis or experiment status changes.

Exact source correspondence

Native SignedGraph::edge calls find(u), then find(v) on the mutated state. It forms cu=su*pu.s and cv=sv*pv.s, then subtracts both signed constant and target offsets. reducedUnionPot and run_refines use that order and connect both decoded finds to the same vertex-value function and target k. The stated bounds retain canonical coefficients, signs plus/minus one, prime positive modulus at most 2^63, valid anchors, a terminating forest and the true incoming equation.

The native forward potential is (-cu*cv, cu*a, cu*b). The backward branch retains its sign and applies -s to each offset. branchWord_refines selects joinWord(ru,rv,p) or joinWord(rv,ru,reversePot(p)); the literal word-potential theorem supplies those exact decoded formulas. The field composition supplies distinct roots and a new forest rank. This is enough for the existing subsequent-find existence theorem; it does not identify the rank with native sizes.

joinWord updates the child parent/potential and transfers an existing child anchor only when the parent lacks one. If both have anchors, it retains both modeled fixed entries. Their validity follows from the incoming truth and prior valid-state assumptions. The native additional constraint return is outside this composition. It is already modeled in the later union-return packet, so it should be integrated rather than reimplemented.

Worked arithmetic witnesses

Over modulus7, choose pu=(-1,2,1), pv=(1,3,2), su=1, sv=-1 and incoming labels(a,b)=(4,5). Reduction gives (cu,cv)=(-1,-1), reduced labels(5,6), and union potential(-1,2,1). With k=3 and rv=4, ru=1; the endpoint values are u=4 and v=6. Both incoming sides equal5 modulo7. A child anchor(1,0) transports to parent anchor(1,1), which evaluates to4 at k=3. A parent anchor(4,0) is compatible at that target even though its affine coefficients differ; its constraint is a separate return obligation.

For sign+1 and offsets(2,3), reversal must use offsets(5,4) modulo7. Copying the old offsets would fail, for example at k=0, parent=0 and child=2. These are reviewer paper calculations, not newly executed Lean controls.

Existing size and return work

The union-size packet already contains nine declarations and seven controls with sixteen retained readbacks. Its source uses Array USize, native <= with ties attaching ru beneath rv, updates only the parent size, and proves exact addition under disjoint-component cardinality premises and n<USize.size. SizeComposition instantiates the reviewed union theorem with precisely that Boolean. No new size-choice theorem is needed.

The union-return packet already models saved pre-update anchors, transport, existing-parent comparison, safe returned target and exclusion of inconsistency under true premises, including two-find composition. Its recorded ten declarations plus seven controls remain producer evidence; this review does not independently revalidate all those proofs.

Smallest remaining size invariant

Define each component as the finite set of vertices whose parent-chain root is r. Require only roots to satisfy size[r].toNat = component(r).card, with parallel vector lengths and n<USize.size. Nonroot size slots are stale by design and must not satisfy a global cardinality equality.

The first small construction is root-map preservation under the existing find/path-compression Run: every vertex retains the same root, hence every root component set is unchanged, and the untouched size vector remains valid. Then establish initialization for the singleton node0; fresh node insertion adds one singleton and extends existing component sets by inclusion; distinct-root attachment replaces the parent component by the disjoint union and leaves other roots unchanged. Feed that disjointness/cardinality result into the existing selected_sum_exact theorem. No new arithmetic or attachment-choice model is needed.

Worked size witness: parent array[0,0,2] has root components{0,1} and{2}, with root sizes2 and1. Merging the smaller root2 beneath root0 yields parent[0,0,0], size[root0]=3, while the old child size[root2]=1 stays unchanged. A specification requiring all size slots to equal their current component size would incorrectly reject this native state. Path compression of[0,0,1] to[0,0,0] preserves the root0 component of size3 without writing any size slot.

The integration must additionally prove array growth/bounds, root distinctness, representation of actual parent chains, exact no-overflow conditions for node count and size addition, and the source dispatcher joining same-root versus distinct-root returns. Operation-counter and allocator semantics remain separate. The true geometric incoming edge is still an explicit premise.

Sources and next action

Read and bind word-union-composition, its literal word-union predecessors, field graph-forest Composition/Union, union-size source/native snapshot and union-return report. The research KB is unavailable; these exact primary sources provide the recorded fallback. No broad repository search, compiler execution, benchmark, candidate proof edit or scientific claim promotion occurred. Coordinator can next authorize the isolated root-map/path-compression invariant construction and arrange the independently bounded replay required for formal review completion.